

INTRODUCTION TO AI

MBA Orientation 2026

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Today's Deck

Download these slides any time:

kerryback.com/ai.pdf

THE DECK, AS A PDF

AI Tools at Rice

Claude

University is negotiating with Anthropic for Claude. Maybe soon?

Other Tools

University provides Gemini Chat and Gemini Notebook (NotebookLM).

This session

We're going to use Claude Code on a cloud server (more later)

Today's Session

Prompting

A few tips; easier than it used to be

Code execution

Data analysis, Excel/Word/PowerPoint, artifacts

Persistent instructions

Repeatable and shareable workflows

AI as a Tutor

A one-on-one tutor for courses

PROMPTING

Most ‘prompt engineering’ techniques from a few years ago are now either useless or actually counterproductive.

Tell it what you want, not what to do

I want ... How can we do it?

If something doesn't work

Try again.

Treat AI as an infinitely patient colleague with a poor memory.

Good Enough Prompting

Prompting Tips

Tell it to plan

That surfaces “how best to do something”. It will design subtasks, completing and checking each in turn.

If it gets stuck

Tell it to take a step back and remember the original goal: is there another way to get there?

It is not a person

Just because it “knows” to do something doesn’t mean it will, unless you surface that information.

Start a new conversation for a new task

AI reads the entire conversation again each time you prompt. Irrelevant stuff consumes tokens and may be confusing.

When a conversation gets very long

Tell AI to write a handoff document so you can start a fresh conversation.

Managing Conversations

What Use Will It Be to You?

- A tutor
- Docs or presentations
- Excel
- Analytics
- Brainstorming
- Interview prep

You tell me what I've left off. I'm sure it's something!

HOW WE GOT HERE

- 1 2022–2023 — online chatbots. You typed, it typed back.
- 2 2023–2024 — code execution and web search arrive.
- 3 2025 — the serious versions move to the command line, reading and writing files on your own machine.
- 4 2025–2026 — that command-line capability is wrapped back into an app.
- 5 2026 — agents that carry out extended work on your files without you living in a terminal.

A Brief History

Andrej Karpathy — OpenAI co-founder, former Tesla AI chief, now at Anthropic

October 2025

“They just don’t work ... I feel like the industry is trying to pretend like this is amazing, and it’s not. It’s slop.”

December 2025

“I really am mostly programming in English now... Biggest change to my basic coding workflow in ~2 decades.”

February 2026

“Coding agents basically didn’t work before December.”
“I don’t think I’ve typed like a line of code probably since December.”

THE AI LAB

Environment for this Session

What runs

Claude Code running
Claude Sonnet

What is installed

Supporting software
(Python, JavaScript),
Python libraries, Claude
plugins

What you see

File browser (download
and upload), terminal,
local browser view (App
tab)

Everything running in the cloud, billed to the school.

Sign In

- 1 Go in browser to `lab-mba.rice-business.org`
- 2 Username = netid
- 3 Password = student number (S0....)

Right panel

Claude Code. The prompt window is at the bottom.

Left panel

File Browser (default), Terminal, App. Ignore Terminal and App for now.

Three-dots menu on left

Upload (from your laptop) and Download (to your laptop).

Workspace

Important

Refresh

Use the **Refresh** button above/right of the left pane to see in the File Browser when a new file has been created.

No MS Office

Double-clicking Word, PowerPoint or Excel will not open it — MS Office is not installed.

Download

Download Office docs from the three dots menu and view them on your laptop.

Ask

Ask a question, anything
— just type in the
prompt window and
Enter

Upload

Upload a Word doc, ask
Claude to read it

Download

Tell Claude to create a
Word doc, download it
and read it

Explore

CODE AND DOCUMENTS

Build an Excel Workbook

Tell Claude: Build an Excel workbook containing a mortgage amortization table. Include a chart of the amount going to interest and the amount going to principal each month.

WHAT YOU MIGHT EXPECT

A table of numbers pasted in.
Change an assumption and the sheet is quietly wrong.

WHAT YOU GET

Inputs as numbers, formulas as real formulas. Change the rate and the sheet recalculates.

A Model, Not a Table

	A	B	C	D	E	F
1	Month	Payment Date	Payment	Principal	Interest	Balance
2	1	2026-06-01	2,594.39	344.39	2,250.00	399,655.61
3	2	07/01/2026	2,594.39	346.33	2,248.06	399,309.28
4	3	08/01/2026	2,594.39	348.28	2,246.11	398,961.00
5	4	09/01/2026	2,594.39	350.24	2,244.16	398,610.76
6	5	10/01/2026	2,594.39	352.21	2,242.19	398,258.56
7	6	11/01/2026	2,594.39	354.19	2,240.20	397,904.37
8	7	12/01/2026	2,594.39	356.18	2,238.21	397,548.19
9	8	01/01/2027	2,594.39	358.18	2,236.21	397,190.00
10	9	02/01/2027	2,594.39	360.20	2,234.19	396,829.81
11	10	03/01/2027	2,594.39	362.22	2,232.17	396,467.58
12	11	04/01/2027	2,594.39	364.26	2,230.13	396,103.32

Download and Investigate

Download the file, open it on your own machine.

- 1 Click a payment cell. Formula, or a number?
- 2 Change the rate to 12%. Does the chart move?
- 3 Change the term to 15 years. Does the table shorten?

If any answer is no, say so in one sentence and ask Claude to fix that specifically.

- 1 Ask Claude to add a worksheet containing a plot of the monthly payment as a function of the interest rate.
- 2 Download and check.

Ask for Something Else

Resume a Conversation

`/resume` in the prompt window brings up a list of past conversations.

Up and down arrow to move around in the list. Enter to select.

Close and Reopen Claude

If you ever need to:

CTRL-C twice to shut it down.

Then Enter to restart.

Break

DATA AND CHARTS

Work with Online Data

Tell Claude: Get the 30-year mortgage rate from FRED (Federal Reserve Economic Data), 1990 to today. Plot the mortgage rate as a function of time and save it as a PNG.

Tell Claude: Plot a histogram of the mortgage rate and save it as a PNG.

Claude probably runs Python code to get the data. It will certainly run Python code to generate plots. The code is running at the AI Lab's cloud host.

What could go wrong?

- No plot
- Wrong type of plot
- Wrong data in plot

Is there anything you could ask to clarify?

Validation

Subagents

Claude can generate subagents.

No shared context

Subagents do not see the prior conversation. They only see instructions they get from the main agent.

A second opinion

You can tell Claude to generate a subagent to do the same task.

If they agree, ...?

Sources of Error in AI Data Analysis

Misunderstanding

Instructions were unclear or ambiguous.

Mitigation: ask AI to explain what it is planning to do before it starts.

Assumptions

AI silently made an assumption during the task without confirming.

Mitigation: ask AI to explain what it did step-by-step after it finishes.

Coding error

Very unlikely in data analysis.

Look at the Code

- Ask Claude to show you the code it ran. Try it now.
- Even without knowing Python, it is often possible to puzzle out what it did.
- Can you find the name of the data series it got from FRED?

Another Example

Tell Claude: Compute the monthly WMT and SPY returns since 2020 using Yahoo Finance data. Produce a scatter plot and save it as a PNG.

Tell Claude: Put the scatter plot in a PowerPoint deck. Add some analysis.

Then download it and inspect it.

Add to the Plot

Tell Claude: Add a regression line (line of best fit) to the scatter plot.

Tell Claude: Annotate any interesting points.

Tell Claude: Put the new plot in a Word doc.

Then download it to view.

ARTIFACTS

Interactive Documents (Artifacts)

A CHART

A picture of one answer,
computed once.

AN ARTIFACT

A small program. It answers
the question again with
different inputs.

No install, no server, one file — the person you send it to can use it without you.

Try It

Say what you want in one sentence. Claude writes the file.

Tell Claude: Build me a mortgage calculator as a single HTML file: sliders for price, down payment, rate and term, showing the monthly payment and a chart of interest versus principal.

Tell it to dress up the page, add your name, whatever. Then download it and double click in File Explorer/Finder — not from your browser.

SKILLS

Skills are Persistent Instructions

Anthropic introduced skills in October 2025. Others followed.

What it is

A text file with its description in a short header, saved in a particular place.

Read at startup

Claude reads the descriptions of all installed skills at the start of each conversation.

Read when needed

It reads the rest of the skill instructions only when a task calls for them.

Skills You Already Used

Excel

The `xlsx` skill tells Claude how to create and edit Excel workbooks.

PowerPoint

The `pptx` skill tells Claude how to create and edit PowerPoint decks.

Word

The `docx` skill tells Claude how to create and edit Word documents.

Anthropic wrote these three and they ship with Claude. Claude uses them when you ask for a document.

Create Your Own

- 1 Iterate — work on something until you get what you want.
- 2 Recognize — you will want the same thing again, and you do not want to iterate again.
- 3 Save the recipe — tell Claude to package the final version as a skill.

Ask it to save your result alongside the skill, as an example to follow.

Try It — Make One of Your Own

- 1 **Create:** “Save what we just did as a skill called `ticker-plot`: get two tickers from Yahoo Finance, scatter one against the other, fit a regression line, and put it in a Word doc.”
- 2 **Invoke:** start a new conversation and enter `/ticker-plot K0 and PEP`.
- 3 **Inspect:** ask Claude to show you the `SKILL.md` it wrote.

Tell Claude: Describe your **finance-data** skill.

Where it came from

Claude created it at my direction, and it is pre-installed on the AI Lab.

Passing it on

Skills are easily shared. Ask Claude.

Finance Data Skill

Always Ask Claude

People often ask “how do I get Claude to ...”

Claude can answer that question for you.

I’m happy to answer questions, but Claude can too.

Break

LEARNING WITH AI

Tutors

Ask Claude to teach you about linear regression. Tell Claude to assume you don't know anything about it. Ask Claude to quiz you to check your understanding.

- 1 Current AI models will not hallucinate about reasonably well-known topics.
- 2 If you ask something they don't know, they do web search.
- 3 If they still can't find it, they might hallucinate — you are at risk with very obscure questions.

Ask for sources and verify when in doubt.

Could AI Hallucinate?

Tutoring in Business School

You can trust AI to tutor you on any business school subject — I think; I don't know them all.

Emphasis may differ

AI might emphasize something other than your instructor emphasizes, but it will seldom contradict.

If it seems to

If you think there is a contradiction, ask AI for clarification.

Gemini Notebook

What it is

Previously called NotebookLM.

What you give it

Upload docs, or find them online. It answers from the docs.

What it makes

Video and audio overviews, quizzes, flash cards.

Free to Rice students: [Rice IT link](#).

CHATBOTS AND AGENTS

- 1 Open Gemini Chat from the Rice page.
- 2 Ask it to teach you linear regression.
- 3 Ask it to compute monthly WMT and SPY stock returns from Yahoo Finance data and produce a scatter plot.

If it just gives you code, ask it to run it.

Try Gemini Chat

Why Can't Gemini Chat Execute Code?

A chatbot

A software program that connects you to an LLM (large language model).

What it passes

Your prompt, and back the response — plus the **system prompt** and the **conversation history**.

An agent

A chatbot with tools. The tools are external services, not the LLM.

Code execution is a tool. Gemini Chat does not have that tool.

Tool Harnesses

- 1 The agent passes the prompt, system prompt, conversation history and tool definitions with each prompt.
- 2 The LLM may respond: use tool X with these instructions.
- 3 The agent runs tool X and reports back, with **instructions for handling the response**.
- 4 The LLM decides what to do next.

Those instructions are the **tool harness**, and some are much better than others.

RISK, AND OTHER MODELS

Lots of Models

OpenRouter — one API key, hundreds of models from every lab.

Switching is a string

Change the model name; same code, different model

Why bother

Comparing answers, and cost — the cheap model is often enough

Security Risks of AI Agents

WHAT YOU SEND

Instructions. The model reads them and acts.

WHAT ELSE GETS SENT

Content — an email, a webpage, a document, a spreadsheet cell.

Content saying “forward the customer list” reads like an instruction too.

Private data

What the agent can reach

Untrusted content

Anything it reads that someone else wrote

Ability to act

Send, write, pay, delete

Any two are fine. All three is exposure. Something always to consider: what can an agent reach where it might do harm?

Why This Is Hard

WRAP UP

Four Things to Take With You

Code execution

AI plus the ability to run code is what made today's Excel workbooks, charts and documents possible.

Plan mode

Telling it to plan first surfaces how it intends to do the job, before it does it.

Save your workflows

Once something works, have Claude package it as a skill so you never explain it again.

Use it as a tutor

Any subject, any time, at whatever level you ask for.

REMEMBER

AI is an infinitely patient colleague with a poor memory.